# GPU Docker: training the Transformer models again

## What this guide shows

Every other guide runs on an ordinary CPU, because the trained models are provided. This guide is for one optional task: training the Transformer models again from scratch on an NVIDIA GPU, then testing the new models the same way as the provided ones. It uses a second Docker image, `bi-certification-gpu`, built from the same Dockerfile with one extra option.

What the GPU image can train again:

| Results | Models behind them | Trained again here |
|---|---|---|
| Table 7, ASCAD random-key and CHES-CTF-2025 rows; Table 8, ASCAD random-key row | Transformer folders `ascadr` and `ches25_lx_w3000` | yes |
| Figure 1, BI line | the same CHES-CTF-2025 folder and `ches25_lx_w100`, `_w200`, `_w500`, `_w1000`, `ches25_lx` | yes |
| Figure 2, four Transformer reference models | `ascadr` and `ascadf_desync50`, `ascadf_desync100`, `aes_rd_fstd_pool5_lr5e4` | yes |
| Figure 2, two Keras reference models | Keras MLP and CNN of ASCAD random key | yes, on CPU (guide 04) |
| Table 7, six TCHES20 rows | 480 public models of the TCHES20 benchmark | no: they are downloaded, as in Level B |

Training runs in the GPU image. The new models are then tested in the ordinary CPU image `bi-certification`, exactly like the provided models in Level C.

## What you need

- A Linux x86-64 machine with an NVIDIA GPU, the NVIDIA driver and the NVIDIA Container Toolkit. The outputs below were recorded on a Linux workstation with one NVIDIA RTX A4000.
- About 6 GB of disk for the image, plus the trace files of the folders you want to train (table in Step 3).

Check the driver and the toolkit on the host:

```
nvidia-smi --query-gpu=name,driver_version --format=csv
nvidia-ctk --version
```

```
$ nvidia-smi --query-gpu=name,driver_version --format=csv && \
    nvidia-ctk --version | head -1
name, driver_version
NVIDIA RTX A4000, 580.178.04
NVIDIA Container Toolkit CLI version 1.20.1
[exit code 0, 0 s]
```

## Step 1: build the GPU image

From the top folder of the repository:

```
docker build --build-arg TF=gpu -t bi-certification-gpu .
```

```
$ docker build --build-arg TF=gpu -t bi-certification-gpu .
[... earlier output omitted; key lines kept ...]
[...]
#9 50.48 Downloading cffi-2.1.1-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (217 kB)
#9 50.48    ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
    ━━━━━━━━ 217.8/217.8 kB 249.4 MB/s eta 0:00:00
#9 50.50 Downloading oauthlib-4.0.0-py3-none-any.whl (159 kB)
#9 50.50    ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
    ━━━━━━━━ 159.7/159.7 kB 660.8 MB/s eta 0:00:00
#9 50.52 Downloading pyasn1-0.6.4-py3-none-any.whl (84 kB)
#9 50.52    ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
    ━━━━━━━━ 84.4/84.4 kB 190.4 MB/s eta 0:00:00
#9 50.54 Downloading pycparser-3.11-py3-none-any.whl (51 kB)
#9 50.54    ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
    ━━━━━━━━ 51.2/51.2 kB 545.3 MB/s eta 0:00:00
#9 51.48 Installing collected packages: mpmath, libclang, flatbuffers, wrapt, urllib3, typing-extensions, te
    rmcolor, tensorflow-io-gcs-filesystem, tensorflow-estimator, tensorboard-data-server, sympy, …
#9 84.14 Successfully installed MarkupSafe-3.0.4 absl-py-2.5.1 astunparse-1.6.3 certifi-2026.7.22 cffi-2.1.1
     charset_normalizer-3.5.2 cryptography-50.0.2 filelock-4.1.1 flatbuffers-25.12.19 fsspec-202 …
#9 84.14 WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour
    with the system package manager. It is recommended to use a virtual environment instead: htt …
#9 DONE 85.9s

#10 [6/6] COPY . .
#10 DONE 0.0s

#11 exporting to image
#11 exporting layers
#11 exporting layers 10.3s done
#11 writing image sha256:3a02d77d52f38e761ff47814aa8d116ef3ea5740525facc48abd81bd7747e666 done
#11 naming to docker.io/library/bi-certification-gpu done
#11 DONE 10.3s

View build details: docker-desktop://dashboard/build/default/default/sllkxckct9xquo7u04h68s1j8
[exit code 0, 1 min 36 s]
```

The option `TF=gpu` installs `requirements-ml-gpu.txt`: TensorFlow 2.15 with its CUDA 12.2 and cuDNN 8.9 libraries. Everything else is the same as in `bi-certification`. Without the option you get the ordinary CPU image.

## Step 2: open a shell in the GPU image

The GPU image shares the same four folders as the CPU image (guide 00). Add `--gpus all` so the container can use the GPU:

```
mkdir -p results datasets/raw checkpoints external
docker run -it --rm --gpus all \
  -v "$PWD/results:/artifact/results" \
  -v "$PWD/datasets/raw:/artifact/datasets/raw" \
  -v "$PWD/checkpoints:/artifact/checkpoints" \
  -v "$PWD/external:/artifact/external" \
  bi-certification-gpu bash
```

Type the commands of the next steps at that prompt. First check that TensorFlow sees the GPU:

```
python -c "import tensorflow as tf; print(tf.config.list_physical_devices('GPU'))"
```

```
$ python -c "import tensorflow as tf; print('TensorFlow', tf.__version__, '| CUDA build:', tf.test.is_built_
    with_cuda(), '| GPUs:', tf.config.list_physical_devices('GPU'))"
TensorFlow 2.15.0 | CUDA build: True | GPUs: [PhysicalDevice(name='/physical_device:GPU:0', device_type='GPU
```

```
    ')]
2026-10-12 14:11:30.073542: E external/local_xla/xla/stream_executor/cuda/cuda_dnn.cc:9261] Unable to regist
    er cuDNN factory: Attempting to register factory for plugin cuDNN when one has already been  …
2026-10-12 14:11:30.073577: E external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:607] Unable to registe
    r cuFFT factory: Attempting to register factory for plugin cuFFT when one has already been r …
2026-10-12 14:11:30.074343: E external/local_xla/xla/stream_executor/cuda/cuda_blas.cc:1515] Unable to regis
    ter cuBLAS factory: Attempting to register factory for plugin cuBLAS when one has already be …
[exit code 0, 3 s]
```

At the start of every GPU command, TensorFlow prints three lines `Unable to register cuDNN/cuFFT/cuBLAS factory`. They are harmless.

## Step 3: get the traces

Each Transformer folder is trained on one trace file:

| Folders | Trace file in `datasets/raw/` | How to get it |
|---|---|---|
| `ascadr` | `ascad_random_key/ASCAD.h5` | `python -m paper_results.fetch_-ascad_random_key` |
| `ascadf_desync50`, `ascadf_desync100` | `tches20/.../ASCAD_desync50.h5`, `ASCAD_desync100.h5` | `paper_results/fetch_tches20.sh` |
| `aes_rd_fstd_pool5_lr5e4` | `aes_rd/AES_RD_estranet_feature_-standardized.h5` | `fetch_tches20.sh`, then `python -m bi_scope_certificates.prepare_-aesrd_estranet_h5` |
| the six `ches25_lx…` folders | `ches_ctf_2025/CHES_Challenge.h5` | download by hand (guide 00) |

The three scripts in one command:

```
python -m paper_results.fetch_ascad_random_key && paper_results/fetch_tches20.sh && \
  python -m bi_scope_certificates.prepare_aesrd_estranet_h5
```

```
$ python -m paper_results.fetch_ascad_random_key && \
    paper_results/fetch_tches20.sh && \
    python -m bi_scope_certificates.prepare_aesrd_estranet_h5
downloading https://www.data.gouv.fr/api/1/datasets/r/b4ace767-c2a4-4db4-8e01-4527b5b91f00
datasets/raw/ascad_random_key/ASCAD.h5: checksum verified
unpacking datasets/raw/tches20/TCHES20V3_CNN_SCA/datasets/ASCAD_dataset.zip
unpacking datasets/raw/tches20/TCHES20V3_CNN_SCA/datasets/AES_HD_dataset.zip
unpacking datasets/raw/tches20/TCHES20V3_CNN_SCA/datasets/AES_RD_dataset.zip
unpacking datasets/raw/tches20/TCHES20V3_CNN_SCA/datasets/DPAv4_dataset.zip
unpacking datasets/raw/tches20/TCHES20V3_CNN_SCA/models/pretrained_models.zip
TCHES20 benchmark ready in datasets/raw/tches20/TCHES20V3_CNN_SCA (480 pretrained models)
wrote=datasets/raw/aes_rd/AES_RD_estranet_feature_standardized.h5
profile=(25000, 3500) attack=(25000, 3500)
trace_mode=feature_standardize target_byte=0
Cloning into 'datasets/raw/tches20/TCHES20V3_CNN_SCA'...
[exit code 0, 31 s]
```

## Step 4: train one Transformer folder

`train_checkpoints.sh` trains one folder with the settings of the paper (README of `bi_suite_8datasets/estranet_-support/`). It downloads the Transformer code into `external/` if it is missing, prints the settings and the number of GPUs it sees, and saves a model every 20,000 steps:

```
bi_suite_8datasets/estranet_support/train_checkpoints.sh ascadr
```

```
$ bi_suite_8datasets/estranet_support/train_checkpoints.sh ascadr
Cloning into 'external/EstraNet'...
EstraNet sources ready in external/EstraNet
suite=ascadr dataset=ASCAD input_length=1400 pool_size=10 beta_hat_2=50 learning_rate=2.5e-4
train_steps=4000000 data=datasets/raw/ascad_random_key/ASCAD.h5 checkpoints=checkpoints/estranet_retrained/a
    scadr gpus=1
2026-10-11 01:41:36.241849: E external/local_xla/xla/stream_executor/cuda/cuda_dnn.cc:9261] Unable to regist
    er cuDNN factory: Attempting to register factory for plugin cuDNN when one has already been  …
2026-10-11 01:41:36.241883: E external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:607] Unable to registe
    r cuFFT factory: Attempting to register factory for plugin cuFFT when one has already been r …
2026-10-11 01:41:36.242608: E external/local_xla/xla/stream_executor/cuda/cuda_blas.cc:1515] Unable to regis
    ter cuBLAS factory: Attempting to register factory for plugin cuBLAS when one has already be …
external/EstraNet/transformer.py:244: SyntaxWarning: "is" with a literal. Did you mean "=="?
  ks = 11 if l is 0 else self.conv_kernel_size
INFO:tensorflow:
[... 838 lines omitted ...]
INFO:tensorflow:[3960000] | gnorm 145.56 lr  0.000001 | loss  4.45
INFO:tensorflow:Train batches[  100]                | loss  4.20
INFO:tensorflow:Eval  batches[  100]                | loss  5.26
INFO:tensorflow:Model saved in path: checkpoints/estranet_retrained/ascadr/trans_long-199
INFO:tensorflow:[3980000] | gnorm 146.48 lr  0.000001 | loss  4.45
INFO:tensorflow:Train batches[  100]                | loss  4.21
INFO:tensorflow:Eval  batches[  100]                | loss  5.32
INFO:tensorflow:Model saved in path: checkpoints/estranet_retrained/ascadr/trans_long-200
INFO:tensorflow:[4000000] | gnorm 146.07 lr  0.000001 | loss  4.45
INFO:tensorflow:Train batches[  100]                | loss  4.19
INFO:tensorflow:Eval  batches[  100]                | loss  5.28
INFO:tensorflow:Model saved in path: checkpoints/estranet_retrained/ascadr/trans_long-201
[exit code 0, 8 h 41 min 6 s]
```

- The new models go to `checkpoints/estranet_retrained/<folder>/trans_long-<n>`, next to the provided ones in `checkpoints/estranet/`. Model 1 is the untrained model, and model `n` has trained for 20,000 × (n – 1) steps; model 201 ends the run.
- If a run stops, the same command continues from the last saved model.
- `gpus=0` at the end of the `train_steps=` line means the container does not see the GPU (check `--gpus all` and Step 2). Training would then run on the CPU, which takes days per folder.

To train all ten folders, one after another, run from the host:

```
for suite in ascadr ascadf_desync50 ascadf_desync100 aes_rd_fstd_pool5_lr5e4 ches25_lx_w100 \
             ches25_lx_w200 ches25_lx_w500 ches25_lx_w1000 ches25_lx_w3000 ches25_lx; do
  docker run --rm --gpus all -v "$PWD/results:/artifact/results" -v "$PWD/datasets/raw:/artifact/datasets/raw" \
    -v "$PWD/checkpoints:/artifact/checkpoints" -v "$PWD/external:/artifact/external" \
    bi-certification-gpu bi_suite_8datasets/estranet_support/train_checkpoints.sh "$suite"
done
```

## Step 5: test the new models

Leave the GPU shell and open the CPU shell of guide 00 (image `bi-certification`, the same four shared folders). The Level C script takes the folder of the new models with `--checkpoint-root`. For the two Table 7 folders:

```
python -m paper_results.evaluate_estranet_suites --suites ascad_random_key \
  --checkpoint-root checkpoints/estranet_retrained --out-dir results/paper_results/retrained
python -m paper_results.evaluate_estranet_suites --suites ches_ctf_2025 \
  --checkpoint-root checkpoints/estranet_retrained --out-dir results/paper_results/retrained
```

```
$ python -m paper_results.evaluate_estranet_suites --suites ascad_random_key --checkpoint-root checkpoints/e
    stranet_retrained --out-dir results/paper_results/retrained
[... earlier output omitted; key lines kept ...]
$ python -m bi_suite_8datasets.evaluate_estranet_bi_suite --project-root external/EstraNet --dataset ASCAD -
    -dataset-name ascad_random_key --scope-id ascad_random_estranet_checkpoint_su …
ascad_random_key_estranet_ckpt1: successes=371 n_te=100000 success_rate=0.00371000 checkpoint=checkpoints/es
    tranet_retrained/ascadr/trans_long-1
ascad_random_key_estranet_ckpt10: successes=434 n_te=100000 success_rate=0.00434000 checkpoint=checkpoints/e
```

```
    stranet_retrained/ascadr/trans_long-10
ascad_random_key_estranet_ckpt20: successes=887 n_te=100000 success_rate=0.00887000 checkpoint=checkpoints/e
    stranet_retrained/ascadr/trans_long-20
ascad_random_key_estranet_ckpt30: successes=1139 n_te=100000 success_rate=0.01139000 checkpoint=checkpoints/
    estranet_retrained/ascadr/trans_long-30
ascad_random_key_estranet_ckpt40: successes=1379 n_te=100000 success_rate=0.01379000 checkpoint=checkpoints/
    estranet_retrained/ascadr/trans_long-40
ascad_random_key_estranet_ckpt50: successes=1447 n_te=100000 success_rate=0.01447000 checkpoint=checkpoints/
    estranet_retrained/ascadr/trans_long-50
ascad_random_key_estranet_ckpt60: successes=1483 n_te=100000 success_rate=0.01483000 checkpoint=checkpoints/
    estranet_retrained/ascadr/trans_long-60
ascad_random_key_estranet_ckpt70: successes=1605 n_te=100000 success_rate=0.01605000 checkpoint=checkpoints/
    estranet_retrained/ascadr/trans_long-70
[...]
6250/6250 [==============================] - ETA: 0s
6250/6250 [==============================] - 44s 7ms/step
ascad_random_key_estranet_ckpt80: successes=1643 n_te=100000 success_rate=0.01643000 checkpoint=checkpoints/
    estranet_retrained/ascadr/trans_long-80

6250/6250 [==============================] - 44s 7ms/step
ascad_random_key_estranet_ckpt90: successes=1664 n_te=100000 success_rate=0.01664000 checkpoint=checkpoints/
    estranet_retrained/ascadr/trans_long-90

6250/6250 [==============================] - 44s 7ms/step
ascad_random_key_estranet_ckpt99: successes=1678 n_te=100000 success_rate=0.01678000 checkpoint=checkpoints/
    estranet_retrained/ascadr/trans_long-99
Wrote results/paper_results/retrained/estranet/ascad_random_key_estranet_success.csv
Wrote results/paper_results/retrained/estranet/ascad_random_key_estranet_success.summary.json
external/EstraNet/transformer.py:244: SyntaxWarning: "is" with a literal. Did you mean "=="?
  ks = 11 if l is 0 else self.conv_kernel_size
site-packages/keras/src/initializers/initializers.py:120: UserWarning: The initializer RandomUniform is unse
    eded and being called multiple times, which will return identical  …
  warnings.warn(
ascad_random_key: checkpoint 99 successes=1678/100000 EPI=0.4989 key recovery at 100/500/1000/2000 traces =
    [97, 100, 100, 100] of 100
[exit code 0, 9 min 29 s]
```

```
$ python -m paper_results.evaluate_estranet_suites --suites ches_ctf_2025 --checkpoint-root checkpoints/estr
    anet_retrained --out-dir results/paper_results/retrained
[... earlier output omitted; key lines kept ...]
$ python -m bi_suite_8datasets.evaluate_estranet_bi_suite --project-root external/EstraNet --dataset CHES25
    --dataset-name ches_ctf_2025 --scope-id ches_ctf_2025_estranet_w3000_checkpoi …
ches_ctf_2025_estranet_ckpt1: successes=425 n_te=100000 success_rate=0.00425000 checkpoint=checkpoints/estra
    net_retrained/ches25_lx_w3000/trans_long-1
ches_ctf_2025_estranet_ckpt2: successes=408 n_te=100000 success_rate=0.00408000 checkpoint=checkpoints/estra
    net_retrained/ches25_lx_w3000/trans_long-2
ches_ctf_2025_estranet_ckpt3: successes=383 n_te=100000 success_rate=0.00383000 checkpoint=checkpoints/estra
    net_retrained/ches25_lx_w3000/trans_long-3
ches_ctf_2025_estranet_ckpt4: successes=374 n_te=100000 success_rate=0.00374000 checkpoint=checkpoints/estra
    net_retrained/ches25_lx_w3000/trans_long-4
ches_ctf_2025_estranet_ckpt5: successes=405 n_te=100000 success_rate=0.00405000 checkpoint=checkpoints/estra
    net_retrained/ches25_lx_w3000/trans_long-5
ches_ctf_2025_estranet_ckpt10: successes=363 n_te=100000 success_rate=0.00363000 checkpoint=checkpoints/estr
    anet_retrained/ches25_lx_w3000/trans_long-10
ches_ctf_2025_estranet_ckpt20: successes=383 n_te=100000 success_rate=0.00383000 checkpoint=checkpoints/estr
    anet_retrained/ches25_lx_w3000/trans_long-20
ches_ctf_2025_estranet_ckpt50: successes=404 n_te=100000 success_rate=0.00404000 checkpoint=checkpoints/estr
    anet_retrained/ches25_lx_w3000/trans_long-50
ches_ctf_2025_estranet_ckpt100: successes=367 n_te=100000 success_rate=0.00367000 checkpoint=checkpoints/est
    ranet_retrained/ches25_lx_w3000/trans_long-100
ches_ctf_2025_estranet_ckpt150: successes=404 n_te=100000 success_rate=0.00404000 checkpoint=checkpoints/est
    ranet_retrained/ches25_lx_w3000/trans_long-150
[...]

6250/6250 [==============================] - 30s 5ms/step
ches_ctf_2025_estranet_ckpt201: successes=417 n_te=100000 success_rate=0.00417000 checkpoint=checkpoints/est
    ranet_retrained/ches25_lx_w3000/trans_long-201
Wrote results/paper_results/retrained/estranet/ches_ctf_2025_estranet_w3000_success.csv
Wrote results/paper_results/retrained/estranet/ches_ctf_2025_estranet_w3000_success.summary.json
external/EstraNet/transformer.py:244: SyntaxWarning: "is" with a literal. Did you mean "=="?
```

```
  ks = 11 if l is 0 else self.conv_kernel_size
site-packages/keras/src/initializers/initializers.py:120: UserWarning: The initializer RandomUniform is unse
    eded and being called multiple times, which will return identical  …
  warnings.warn(
[CHES25] Loaded LucianoXu drift stats from datasets/raw/ches_ctf_2025/CHES_Challenge.h5.estranet_lucianoxu_d
    n1_len3000_prof450000_att100000.npz
[CHES25] Available keys in 'Attack_traces': ['metadata', 'traces']
[CHES25] Loaded 100000 traces of shape (100000, 3000)
[CHES25] Loaded LucianoXu drift stats from datasets/raw/ches_ctf_2025/CHES_Challenge.h5.estranet_lucianoxu_d
    n1_len3000_prof450000_att100000.npz
[CHES25] Applied LucianoXu drift-fix preprocessing to test traces
[CHES25] Using labels from structured metadata
ches_ctf_2025: checkpoint 1 successes=425/100000 EPI=0.0000 key recovery at 100/500/1000/2000 traces = [0, 0
    , 0, 0] of 100
[exit code 0, 6 min 33 s]
```

Each test prints the correct guesses of the eleven models of the folder, then EPI and key recovery of the model with the most correct guesses, as in Level C. Table 7 from these counts:

```
python -m paper_results.table7_bi_suite --recomputed-dir results/paper_results/retrained \
  --out-dir results/paper_results/retrained_table7
cat results/paper_results/retrained_table7/table7_bi_suite.md
```

```
$ python -m paper_results.table7_bi_suite --recomputed-dir results/paper_results/retrained --out-dir results
    /paper_results/retrained_table7 >/dev/null 2>&1; cat results/paper_results/retrained_table7/table7_bi_su
    ite.md
### Table 7: finite-suite certificates (delta = 1e-06)

| Dataset | d | M | n_te | p_obs | [p-, p+] | BI_obs | [BI-, BI+] | R_margin | EPI | counts |
|---|---|---|---|---|---|---|---|---|---|---|
| ASCAD d0 | 700 | 80 | 10,000 | 0.00630 | [0.00260, 0.01247] | 0.690 | [0.000, 1.675] | 2.58 | -0.000 | arc
    hived |
| ASCAD d50 | 700 | 80 | 10,000 | 0.00970 | [0.00485, 0.01701] | 1.312 | [0.311, 2.123] | 1.26 | 0.239 | arc
    hived |
| ASCAD d100 | 700 | 80 | 10,000 | 0.00980 | [0.00492, 0.01714] | 1.327 | [0.332, 2.134] | 1.25 | 0.208 | ar
    chived |
| ASCAD random | 1,400 | 11 | 100,000 | 0.01678 | [0.01453, 0.01925] | 2.103 | [1.895, 2.301] | 0.19 | 0.499
     | recomputed |
| AES-HD | 1,250 | 80 | 25,000 | 0.00520 | [0.00288, 0.00851] | 0.413 | [0.000, 1.124] | 2.56 | -0.006 | arc
    hived |
| AES-RD | 3,500 | 80 | 12,500 | 0.13968 | [0.12135, 0.15946] | 5.160 | [4.957, 5.351] | 0.15 | 3.097 | arch
    ived |
| DPAv4 | 4,000 | 80 | 500 | 0.02400 | [0.00201, 0.09144] | 2.619 | [0.000, 4.549] | 3.36 | 2.251 | archived
     |
| CHES-CTF-2025 | 3,000 | 11 | 100,000 | 0.00425 | [0.00316, 0.00556] | 0.122 | [0.000, 0.509] | 3.81 | 0.00
    0 | recomputed |
[exit code 0, 1 s]
```

The two rows marked `recomputed` come from the new models; the six TCHES20 rows keep their archived counts. For ASCAD random key the new models give a certified bound BI+ of 2.301 bits (2.313 with the provided models), and for CHES-CTF-2025 0.509 bits (0.521).

For the window folders of Figure 1 add `--suites ches_windows`. For the Figure 2 reference models, pass the same `--checkpoint-root` to `bi_scope_certificates.profile_epsilon_calibration`.

## How long each step takes

Measured on the Linux workstation with one NVIDIA RTX A4000. A full run is 4,000,000 training steps (AES-RD 200,000).

| Step | Image | Time |
|---|---|---|
| build the GPU image (first time) | GPU | 1.5 minutes |

| Step | Image | Time |
|---|---|---|
| get the traces (Step 3) | GPU | 30 seconds, plus the downloads |
| train `ascadr` (Table 7, Table 8, Figure 2) | GPU | 8 hours 41 minutes |
| train `ches25_lx_w3000` (Table 7, Figure 1) | GPU | 7 hours 12 minutes |
| train `ascadf_desync50` (Figure 2) | GPU | about 10 hours 40 minutes |
| train `ascadf_desync100` (Figure 2) | GPU | about 10 hours 40 minutes |
| test the new `ascadr` models | CPU | 9.5 minutes |
| test the new `ches25_lx_w3000` models | CPU | 6.5 minutes |
| Table 7 from the new counts | CPU | 1 second |

The two “about” times come from the speed over 40,000 steps; for `ascadr` the same method gives 8 hours 38 minutes, against 8 hours 41 minutes for the full run. The times of the remaining six folders will be added here.

## What to look for

- `gpus=1` at the end of the `train_steps=` line of every training run.
- Twenty more saved models per 400,000 steps in `checkpoints/estranet_retrained/<folder>/`.
- The test of the new models prints `successes=…` for each of the eleven models of a folder, and the selected model’s EPI and key recovery, as in Level C.